

Report Period	Sep 26, 2026 20:32 - Sep 26, 2026 20:39
Generated At	2026-09-26 20:39:21
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		26537	
Total Threats Mitigated	26	Total Packets Analyzed		3196995599	
High Priority	2	Malicious Packets		3189656541	
Low Priority	24	True Negatives Passed		20762	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		18	
False Positives	99	UDP Flood		2	
FP Rate	0.32%	Uncertain		6	
IF Precision	98.86%				
IF Recall	95.98%	Action		Count	
IF F1-Score	97.40%	Quarantined		4	
IF Accuracy	98.85%	Time Ban		4	
		Blackhole		0	
		Released		3	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	98.86%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.98%	Share of actual attacks successfully flagged
F1-Score	97.40%	Balanced measure combining Precision and Recall
Accuracy	98.85%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.32%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.02%	Actual attacks that went undetected
True Positive Rate (TPR)	95.98%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.68%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5709	FN 239
Actual: Normal	FP 66	TN 20523

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.95%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.97%	Balanced measure combining Precision and Recall
Accuracy	99.95%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	4.55%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	95.45%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	5709	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	98.86%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.98%	Share of actual attacks successfully flagged
F1-Score	97.40%	Balanced measure combining Precision and Recall
Accuracy	98.85%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.32%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.02%	Actual attacks that went undetected
True Positive Rate (TPR)	95.98%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.68%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 5709	FN 239
Actual: Normal	FP 66	TN 20523

2d. Response Latency

Metric	Value	Description
Detection Time	492.6 ms	Average time taken to detect an attack
Mitigation Response Time	1.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.14%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	16.48%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	12.88%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	5	4	2026-09-26 20:36:15	2026-09-26 20:38:56
10.0.0.17	4	3	2026-09-26 20:36:25	2026-09-26 20:39:08
10.0.0.18	4	3	2026-09-26 20:36:14	2026-09-26 20:38:55
10.0.0.19	4	3	2026-09-26 20:36:15	2026-09-26 20:38:56
10.0.0.8	2	1	2026-09-26 20:37:40	2026-09-26 20:38:53
10.0.0.5	3	1	2026-09-26 20:36:43	2026-09-26 20:37:14

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 20:36:14	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:36:15	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:36:15	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:36:25	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:36:25	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 20:36:25	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 20:36:25	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 20:36:26	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban (30s)
2026-09-26 20:36:26	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-26 20:36:27	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 20:36:35	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (30s)
2026-09-26 20:36:36	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-26 20:36:53	10.0.0.5	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 20:36:55	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-26 20:37:05	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Time Ban (1m)
2026-09-26 20:37:26	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-26 20:37:27	10.0.0.19	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-26 20:37:36	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (1m)
2026-09-26 20:37:51	10.0.0.8	DDoS	Uncertain	68.7%	Low	Released
2026-09-26 20:38:06	10.0.0.16	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-26 20:38:23	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 20:38:55	10.0.0.18	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-26 20:38:56	10.0.0.19	DDoS	SYN Flood	90.9%	Critical	Time Ban (2m)
2026-09-26 20:38:56	10.0.0.16	DDoS	SYN Flood	90.9%	Critical	Time Ban (5m)
2026-09-26 20:38:57	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 20:39:08	10.0.0.17	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6425	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:39:08	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6347	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:38:56	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6521	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:38:56	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6467	93.4%	Critical	Phase 2	1	11s	2026-09-26 20:38:55	Attack persisted - escalating to time ban
10.0.0.8	UDP Flood	0.5805	92.4%	High	Phase 2	1	30s	2026-09-26 20:38:53	Ban Expired
10.0.0.16	SYN Flood	0.6370	94.0%	Critical	Phase 2	4	1m 11s	2026-09-26 20:38:06	Attack persisted - escalating to time ban
10.0.0.8	Uncertain	0.6214	68.7%	Low	Phase 1	1	11s	2026-09-26 20:37:51	Attack Stopped
10.0.0.17	SYN Flood	0.6273	93.1%	Critical	Phase 2	3	1m 11s	2026-09-26 20:37:36	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6523	93.3%	Critical	Phase 2	3	1m 12s	2026-09-26 20:37:27	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6381	92.9%	Critical	Phase 2	3	1m 12s	2026-09-26 20:37:26	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6146	58.9%	Low	Phase 1	1	11s	2026-09-26 20:37:14	Attack Stopped
10.0.0.16	SYN Flood	0.6386	94.0%	Critical	Phase 2	3	10s	2026-09-26 20:37:05	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6001	93.1%	High	Phase 2	2	40s	2026-09-26 20:37:05	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6146	58.9%	Low	Phase 1	1	10s	2026-09-26 20:37:03	Attack Stopped
10.0.0.19	SYN Flood	0.6152	93.3%	Critical	Phase 2	2	41s	2026-09-26 20:36:56	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6174	94.0%	Critical	Phase 2	1	40s	2026-09-26 20:36:55	Ban Expired
10.0.0.18	SYN Flood	0.6421	92.9%	Critical	Phase 2	2	41s	2026-09-26 20:36:55	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6130	58.9%	Low	Phase 1	1	10s	2026-09-26 20:36:53	Attack Stopped
10.0.0.17	SYN Flood	0.6355	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:36:35	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6473	91.4%	Critical	Phase 2	1	11s	2026-09-26 20:36:26	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6360	92.9%	Critical	Phase 2	1	10s	2026-09-26 20:36:25	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6478	92.9%	Critical	Phase 2	1	11s	2026-09-26 20:36:25	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME
Network Administrator

DATE